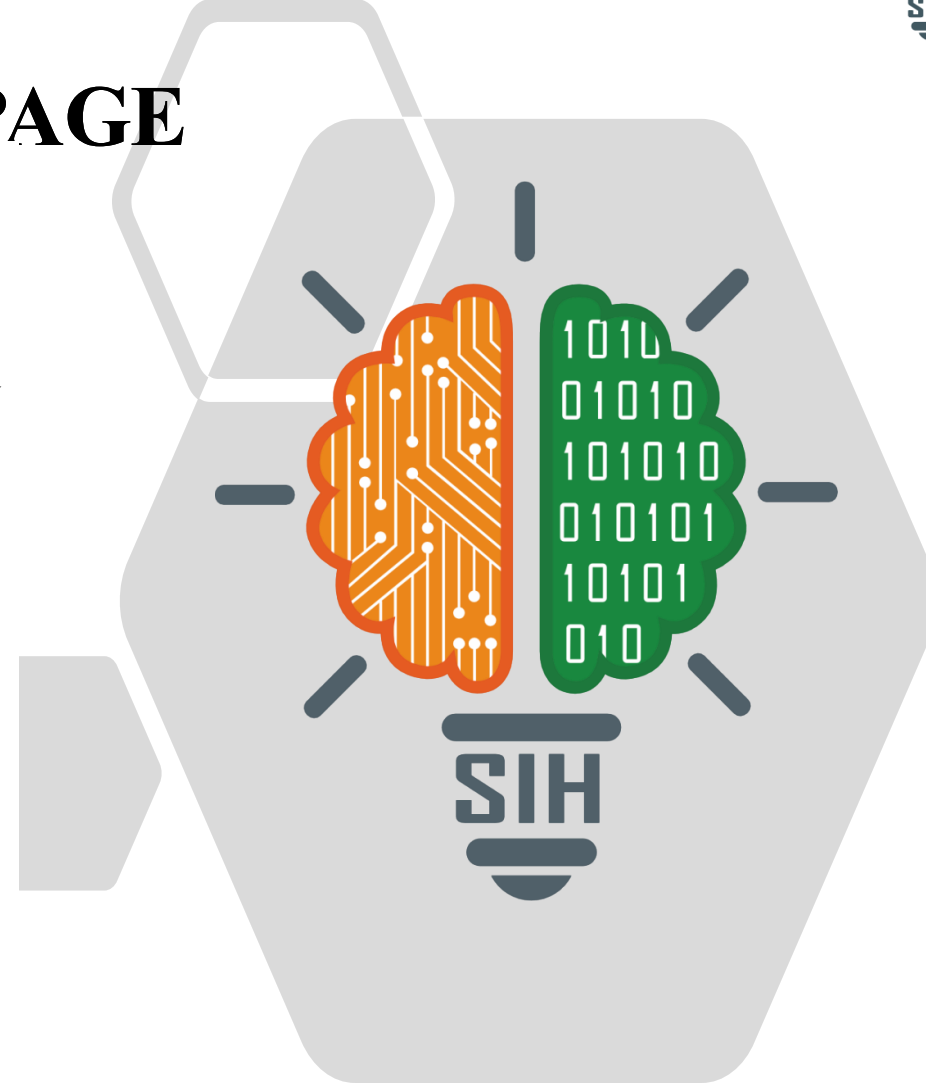


TITLE PAGE

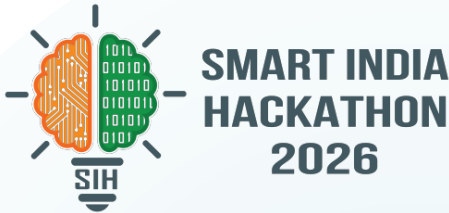
- Problem Statement ID – SIH26153
- Problem Statement Title- *"AI Based Network Attack Forecasting Using Network Traffic Data"*
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID- SIH-A0H-T379
- Team Name (Registered on portal) - TechTonic





AI-Based Network Attack Forecasting

Proposed Solution



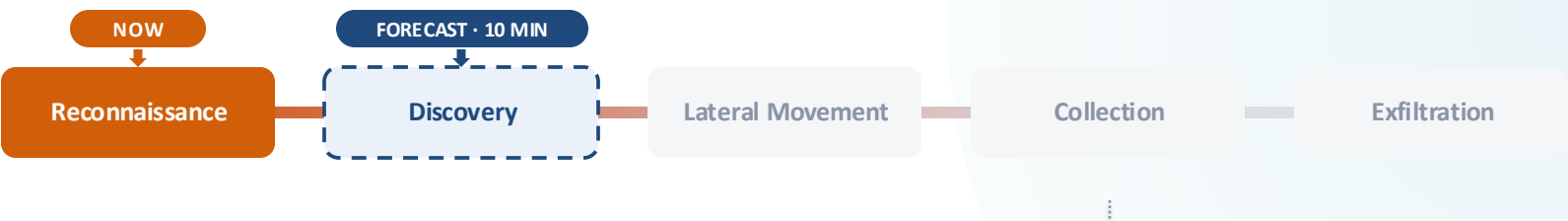
WHY TODAY'S DEFENCE ARRIVES LATE

- 1 Thresholds fire on volume.**
Scanning and discovery traffic is small. Low-and-slow behaviour stays under every rate rule.
- 2 The alert describes the present.**
By the time it fires, the stage it names has already completed.
- 3 A score is not a decision.**
The analyst gets a number — no stage, no reason, no deadline.



We do not detect the attack. We forecast the next ATT&CK stage from flow metadata — and state how much time is left.

ATT&CK TACTICS FOLLOW A TYPICAL PROGRESSION — THE NEXT STEP IN IT IS WHAT WE FORECAST



THE ANALYST CARD

MOCK-UP · NOTHING MEASURED

HOST	10.0.14.7 · window 14:05–14:20
OBSERVED	38 distinct destination ports · 12 new peers · 0 of 38 auths succeeded
NOW	Reconnaissance (TA0043) — scanning behaviour
NEXT	Discovery (TA0007) — T1046 Network Service Discovery · p = 0.62
WHY	host-degree x4 in one window · destination spread ↑ · auth failures ↑

→ in this window, about 10 minutes of usable warning before the next stage

STAGE-AWARE
the output is a named ATT&CK tactic, not a score

CALIBRATED
a stated 0.62 has to behave like 0.62 in replay

TIME-BOXED
every alert carries the warning time still left

AI-Based Network Attack Forecasting

Technical Approach

01 COLLECT

- Zeek conn.log, ssl.log, dns.log
- Connection, TLS-handshake and DNS metadata — never payload
- No decryption, no interception
- Public ATT&CK-labelled corpora

02 REPRESENT

- 15-min rolling window per host
- Temporal view: rate, duration, port spread, burstiness
- Structural view: host = node, connection = edge

03 FORECAST

- Fast path — gradient-boosted trees over the window features
- Slow path — inductive graph embedding on the host graph
- Gated fusion → P(next tactic) over 5 classes

04 EXPLAIN & ACT

- Isotonic calibration, per stage
- Map to ATT&CK tactic + technique
- Feature contributions per alert
- Lead time to the next stage

RELIABILITY LAYER

Drift monitor on feature distributions

Load shedding under burst

Fallback to fast path alone

Immutable audit log per alert

THE LABEL BOTTLENECK

UWF-ZeekData22 is the only corpus labelled by ATT&CK tactic. Its class counts decide what we can train first:

Reconnaissance	9,278,722	OK
Discovery	2,086	OK
Lateral Movement	4	THIN
Persistence	1	THIN

Phase 1 trains the Reconnaissance → Discovery transition. Later transitions need a documented family → tactic mapping over CICIDS2017 and CTU-13 — the project's main risk.

INPUT → OUTPUT CONTRACT

IN one 15-min host-flow window

OUT P(next tactic) over 5 classes + lead time

Discovery		0.62
No escalation		0.24
Lateral Movement		0.08
Collection		0.04
Exfiltration		0.02

Escalation is rare per window, so calibration is scored against the base rate, not accuracy.

REJECTED: PAYLOAD INSPECTION

We started here and dropped it.

Most traffic is TLS-encrypted, so deep inspection means interception — which raises the privacy, legal and storage cost of every deployment.

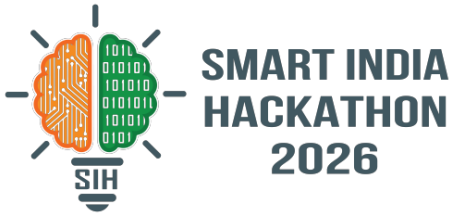
Anderson & McGrew (KDD 2017) separate malicious traffic using connection, TLS-handshake and DNS metadata, without decrypting anything.

So we read the handshake, never the contents.

→ Metadata-only is a commitment, not a limitation.



AI-Based Network Attack Forecasting Feasibility & Viability



WHAT WE COMMIT TO, AND WHY IT HOLDS

DIMENSION	OUR COMMITMENT	WHY IT IS FEASIBLE
Input	Connection, TLS-handshake and DNS metadata	Survives encryption; no interception, far less storage and privacy exposure
Window	15-min rolling window, 5-min stride, per host	Bounds per-host state; the 10-min horizon is a training choice, to be validated
Compute	CPU-first; the tree baseline trains on a laptop	A GPU is needed only for the graph path in Phase 3
Labels	UWF-ZeekData22/24 first; CICIDS2017, CTU-13, LANL after mapping	Only UWF carries ATT&CK tactic labels — the rest need a mapping we must document
Deployment	On-premise, air-gap capable	No cloud dependency anywhere in the scoring path

SCALE-OUT BOUNDARY — FOUR DECISIONS MADE UP FRONT

Stateless scoring workers

Windows partitioned by host hash

Versioned model artifacts

Per-environment recalibration

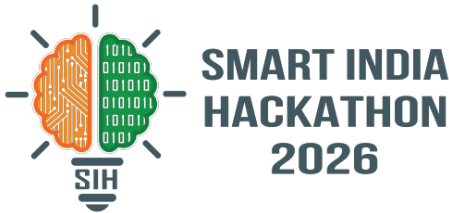
WHAT THIS SYSTEM WILL NOT DO

- Read or classify attack content — there is no payload to inspect.
- Catch an attacker who stays inside normal volume and timing for a whole window.
- Forecast more than one step — we predict the next tactic, not the kill chain.
- Work anywhere flow logs are not already being collected.

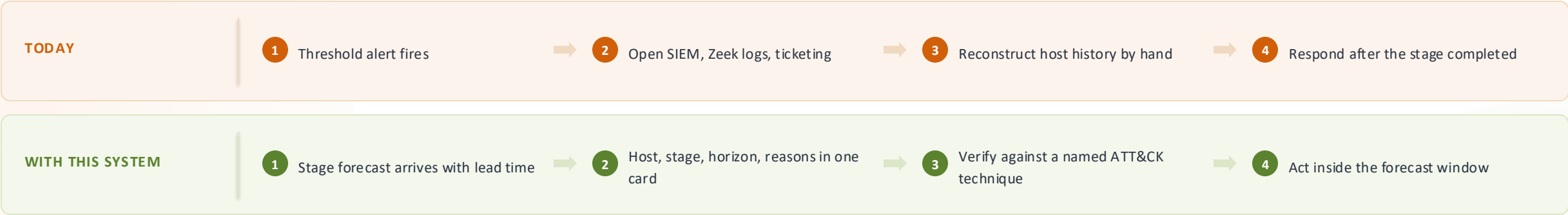


AI-Based Network Attack Forecasting

Impact & Benefits



WHAT CHANGES FOR THE ANALYST ON SHIFT



THREE TARGETS — AND HOW EACH ONE GETS MEASURED

LEAD TIME

TARGET

≥ 5 min median warning before the next stage begins

HOW

(first labelled next-stage event) – (forecast timestamp), on held-out replay

ALERT BURDEN

TARGET

≥ 30% fewer alerts than a fixed-rate threshold detector

HOW

alerts per hour at matched recall, UWF-ZeekData22 replay

TRIAGE EFFORT

TARGET

Zero tool switches needed to understand an alert

HOW

stage, probability, horizon and top-3 reasons present on every card

INTENDED DEPLOYMENT CONTEXT

Campus / enterprise SOC

the shift analyst gets a next-stage forecast for internal hosts

ISP or telecom NOC

flow records already exist at the router — nothing new to capture

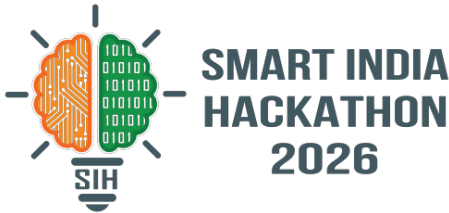
Regulated or air-gapped networks

metadata-only scoring runs entirely on premise



AI-Based Network Attack Forecasting

Research & References



FIVE SOURCES — EACH ONE DECIDES SOMETHING IN THE DESIGN

SOURCE	WHAT IT ESTABLISHES	WHERE IT LANDS IN OUR DESIGN
Freeman, Bagui, Bagui, Mink, Cameron & De Carvalho (2026) Electronics 15(4):871 · doi 10.3390/electronics15040871	ARIMA forecasts traffic to surface anomalies; a decision tree then labels the tactic already present — gains on Discovery, Persistence, Privilege Escalation and Defense Evasion	Confirms tactics are inferable from Zeek logs, and marks the gap we aim at: it labels the present, we forecast the next
Anderson & McGrew (2017) KDD '17 · encrypted malware traffic classification	Malicious traffic is separable using connection, TLS-handshake and DNS metadata, with no decryption	Fixes our input contract at conn / ssl / dns logs — and is why payload inspection was rejected
Hamilton, Ying & Leskovec (2017) NeurIPS '17 · GraphSAGE · arXiv 1706.02216	Inductive node embeddings generalise to nodes that were never seen during training	The slow path has to score hosts that were absent from the training network
Guo, Pleiss, Sun & Weinberger (2017) ICML '17 · arXiv 1706.04599	Classifiers are systematically miscalibrated, and cheap post-hoc scaling corrects it	Why a calibration layer sits between model and analyst — isotonic, since our baseline is trees
MITRE ATT&CK TA0043 · TA0007 · TA0008 · technique T1046	A curated taxonomy of adversary tactics, with a typical order of presentation	Supplies the label space, and the progression our forecast is defined over

ALL LINKS VERIFIED mdpi.com/2079-9292/15/4/871 · arxiv.org/abs/1706.02216 · arxiv.org/abs/1706.04599 · datasets.uwf.edu · unb.ca/cic/datasets/ids-2017.html · csr.lanl.gov/data · stratosphereips.org/datasets-ctu13 · attack.mitre.org

EVALUATION CORPORA

UWF-ZeekData22/24 9.28 M Zeek records, labelled by ATT&CK tactic	CICIDS2017 5 days, 4 with attacks · 8 families · 80+ features	CTU-13 13 labelled botnet scenarios · bidirectional NetFlow	LANL 58 days · 1.65 B events · red-team ground truth
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OUR CONTRIBUTION Prior work forecasts traffic and labels the tactic already present. We forecast the **next tactic itself** — calibrated, explained, with a lead time.